V. Kannnan
École Polytechnique

Open research position (Bachelor)

The Mechanics and Materials under eXtremes research group within the Laboratoire de Mécanique des Solides (LMS) at the Ecole Polytechnique in Palaiseau, France, is looking for passionate **undergraduate** students to study the **dynamic behavior of materials using a micro-Kolsky bar**. The project is expected to start between **January 11-18, 2027** for a duration of **8 weeks**.

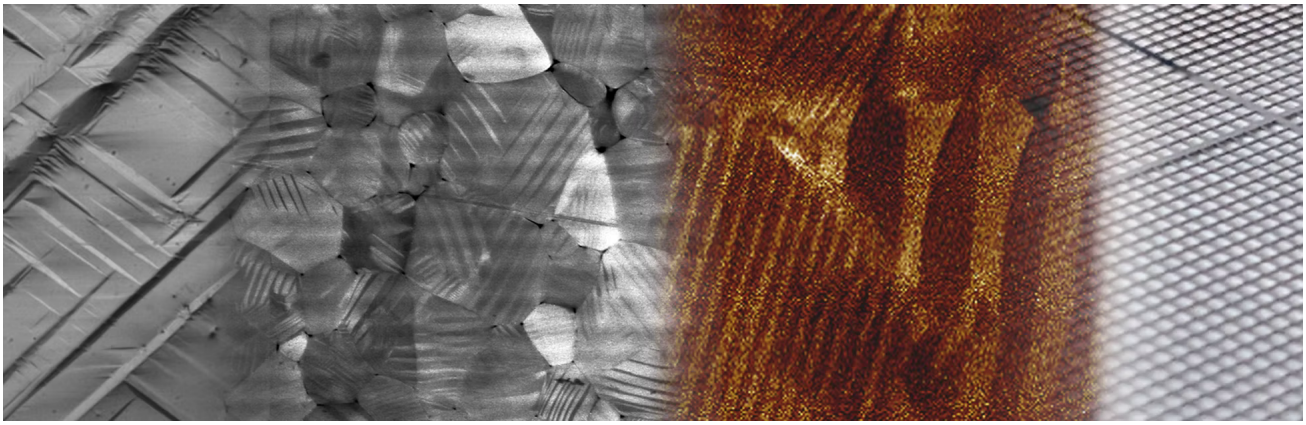
Research topic

Context: The constitutive stress-strain behavior of materials are strongly dependent on the strain rate and stress state of loading. Measurement of the dynamic response, and the fundamental mechanisms governing material behavior such extreme dynamic conditions (withing time scales of microseconds) are long standing challenges in mechanics of materials.
Research project: The student will focus on the development of experimental protocol, high-speed optical instrumentation, and in-situ imaging for a new high-precision "micro-" Kolsky bar experiment developed within the group. The experiment is designed to push the maximum achievable strain rates up to 10^5/s in hundreds of micrometer samples (deformation within tens of microseconds). The student will perform first experiments on copper at these very high strain rates, perform data processing to extract dynamic stress-strain data, and compare with state of the art in literature.

Your profile

- Third year undergraduate from the Math/Physics/Mechanics track of l'X.
- Background in mechanics and/or materials (demonstrated through coursework/internships)
- Eager to work on experimental development including mechanical testing and instrumentation
- Experience with data processing tools using Python/Matlab or equivalent
- Comfortable communicating in English, and enjoy working in an international environment.

Finally and most importantly, if you are passionate and eager to study mechanics, develop rigorous experiments, and enjoy spending many hours in an experimental lab, you are eligible to apply!



For further details contact,
Prof. Vignesh Kannan
Department of Mechanics, Laboratoire de Mécanique des Solides
Pôle Mécanique, [Office] 104: 20-65, [Email] vignesh.kannan@polytechnique.edu

About our team – Mechanics and Materials under eXtremes

The mXm team, led by Prof. Vignesh Kannan, studies the mechanics of materials and structures under large stresses, multi-physical fields and short time scales. We use mechanical design, electronic and optical instrumentation to push the cutting edge of experimental mechanics down to nano-second time scales, and μm length scales. Our experiments are motivated by theoretical foundations, with the goal of understanding fundamental mechanisms that govern the multi-physical response of materials. We continuously seek to develop a collaborative, supportive and international group of scientists who enjoy the pursuit of exciting problems, and each other's company!

To apply!

Please send an email to Prof. Vignesh Kannan (vignesh.kannan@polytechnique.edu) with your CV and transcript with the subject **"Internship application – Bachelor 2027"**.